Trần Quốc Cường

Pentest Internship

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OBJECTIVE

Motivated cybersecurity student seeking a penetration testing internship with a focus on Red Team activities, aiming to gain practical experience in vulnerability assessment and attack simulation.

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT) 2023 - Now

Information Security

GPA: 3.34/4.0 (Awarded academic scholarship) [Information Security Fundamentals - A, Computer Networks - B+, ...]

TECHNICAL SKILLS

Web & App Pentesting	Use tools like Nmap, Gobuster, Hydra, Fuzz and Burp Suite, Wireshark,...
Languages	C, C++, Java, Javascript, Python, PHP (basic syntax)
System & Network	Linux, Windows and Windows Server (AD setup), TCP/IP, HTTP/HTTPS, DNS, OSWAP top 10
Security Basics	Cryptography (SHA, JWT, HMAC, ...),
Experience	Practice on TryHackMe, picoCTF, HackTheBox, Portswigger Lab, Rootme

CERTIFICATIONS

APTIS ESOL (British Council) – B2 Level (CEFR)	3/2024
Cybersecurity Foundations (IBM)	2024
Linux Introduction (Coursera)	2024
Penetration testing – A hand-on approach (PTIT&SamSung)	10/2025

PROJECTS

Linux Log Anomaly Detection 03/2026 - 04/2026

Python, Pandas, Scikit-learn, Linux

- Built a real-time log analysis tool to detect anomaly on Linux systems using /var/log/auth.log.
- Parsed and normalized log data (timestamp, IP, username, status, port) into structured format (CSV).
- Engineered behavioral features like: failed login attempts (ex: 1m), unique IP/user count, login frequency & time-based patterns
- Applied Isolation Forest to detect anomalous login behavior.
- Simulated attacks using Hydra (Kali Linux) to validate detection capability.

Github: https://github.com/lucastran05/linux-log-anomaly-detection-PYP

CVE-2023-48223 – Algorithm Confusion Exploit Analysis 02/2026 - 03/2026

Node.js, JWT, Express, fast-jwt

- Conducted in-depth analysis of CVE-2023-48223 affecting JWT verification in the fast-jwt library.
- Demonstrated an algorithm confusion attack enabling authentication bypass and privilege escalation.

- Forged malicious HS256 tokens signed with RSA public keys to bypass signature verification.
- Investigated library internals to trace flawed key detection and verification logic.
- Recommended secure implementation practices including explicit algorithm whitelisting.

Link: https://hackmd.io/@a3oVEmssREaHsbt5Z_FZmQ/SJwL5Q_i-x

ACHIEVEMENTS

Top 20/160 (Honorable Mention) teams PTIT CTF 2025 (Capture The Flag)	9/2025
Top 8 team (Honorable Mention) P-Innovation 2025 (Innovative Product Competition)	4/2025